



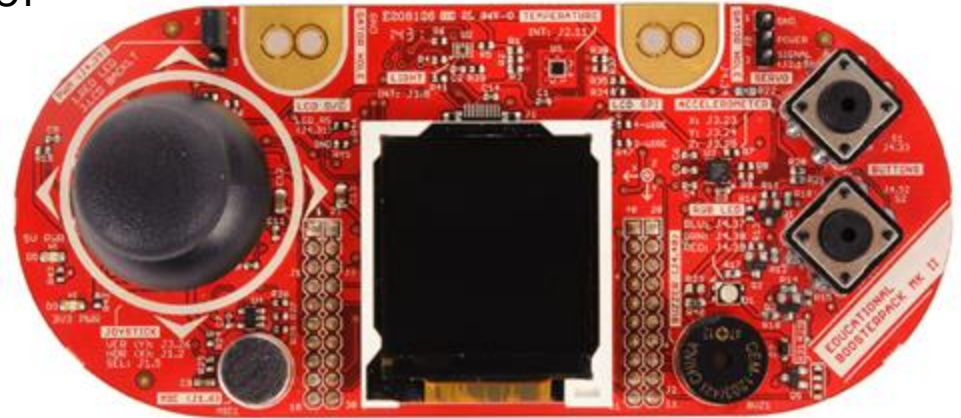
Embedded Software for the Internet Of Things

MSP432 – BOOSTERPACK SENSORS

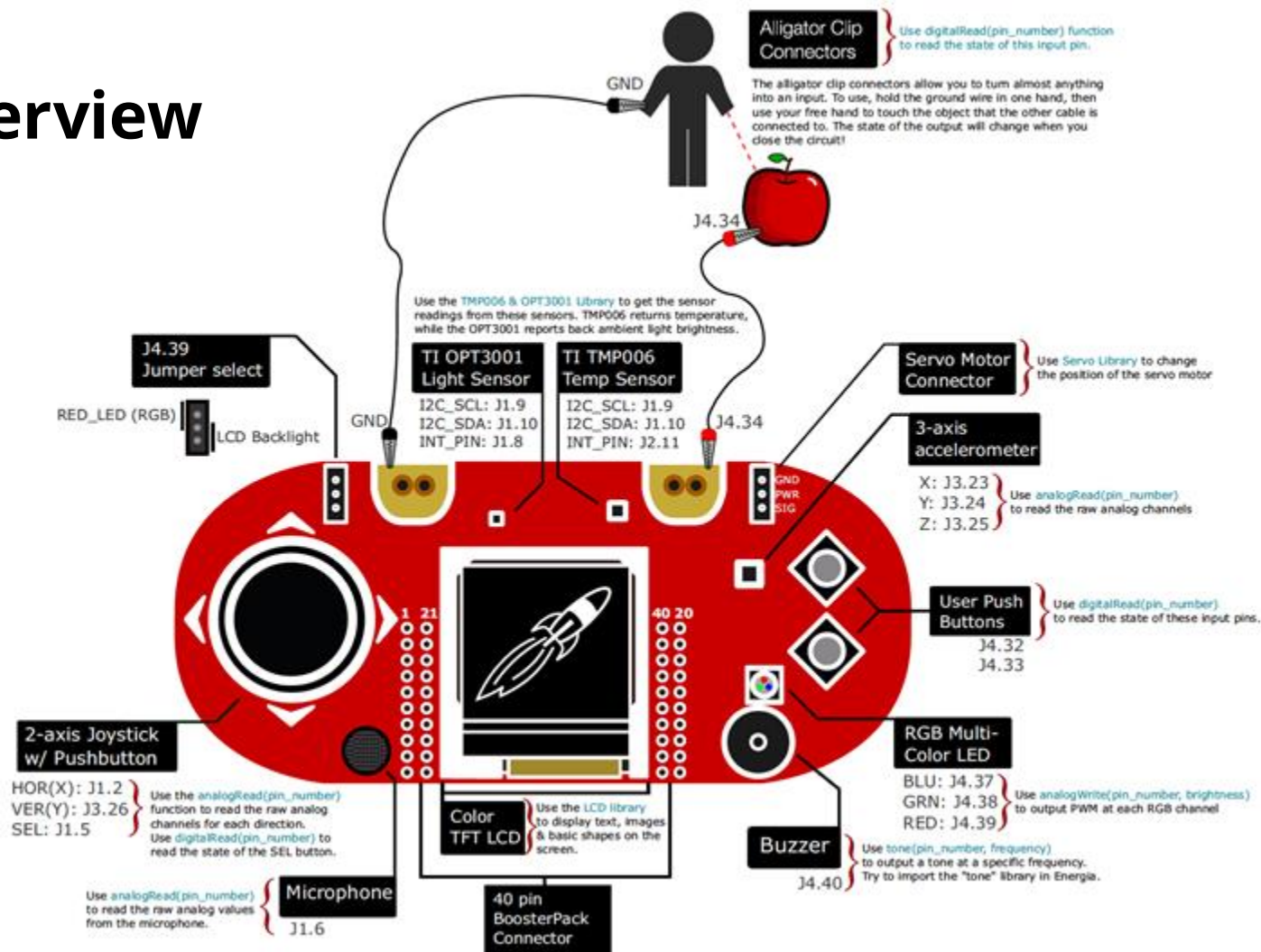
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University of Trento, Italy

BOOSTXL-EDUMKII EDUCATIONAL BOOSTERPACK

- Various analog and digital inputs and outputs
- TI OPT3001 Light Sensor
- TI TMP006 Temperature Sensor
- Servo Motor Connector
- 3-Axis Accelerometer
- User Push Buttons
- RGB Multi-color LED
- Buzzer
- Color TFT LCD Display
- Microphone
- 2-Axis Joystick with Pushbutton



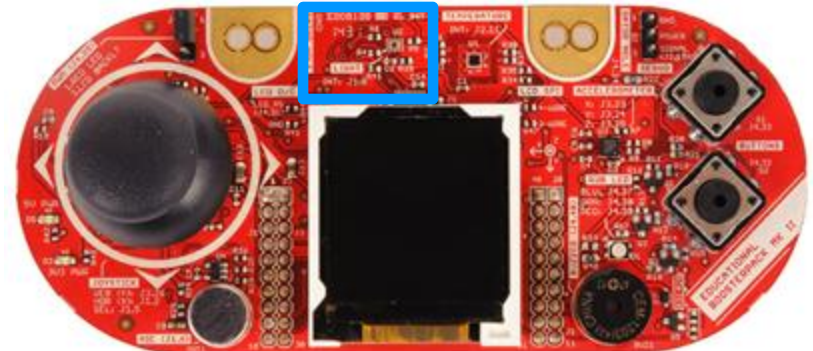
Sensor Overview



TI OPT3001 LIGHT SENSOR

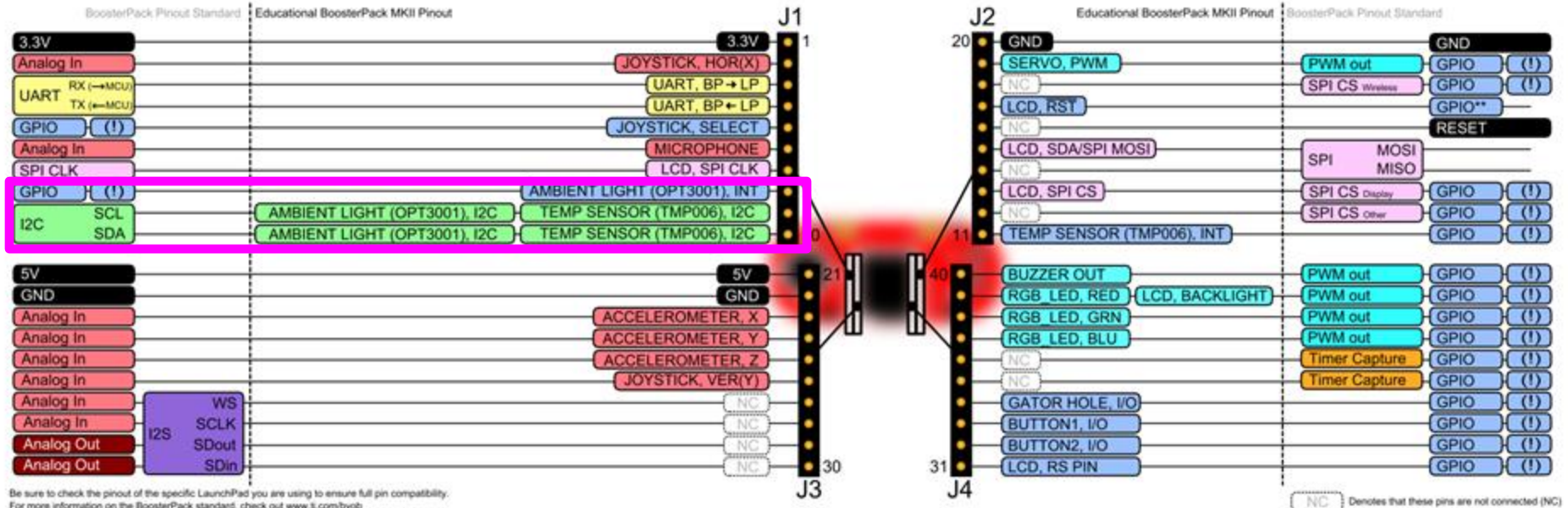
- The OPT3001 is a digital **ambient light sensor (ALS)** that measures the **intensity** of light as visible by the human eye.
 - Covering the sensor with your finger or shining a flashlight on it will change the output of the OPT3001.

BoosterPack Plug-in Module Header Connection	Pin Function
J1.8	OPT3001 interrupt
J1.9 ⁽¹⁾	I ² C SCL
J1.10 ⁽¹⁾	I ² C SDA



PinOut Diagram

- Pins **6.5** (SCL) and **6.4** (SDA)



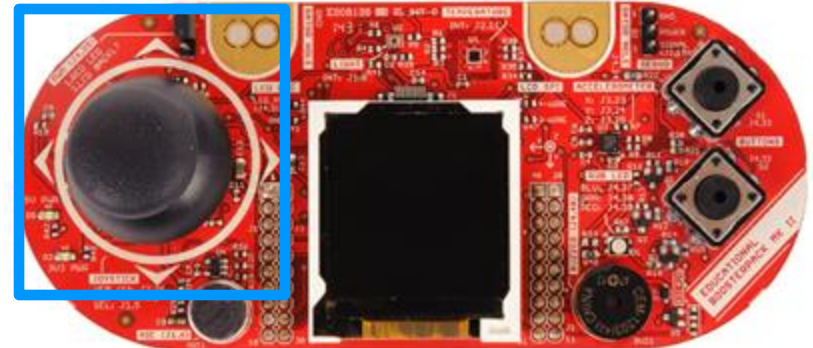
Lab#1: Occupancy Detection

- Try to implement an “**occupancy detection**” algorithm using the Light Sensor.
 - If you detect occupancy, then toggle LED!

2-Axis Joystick with Push Button

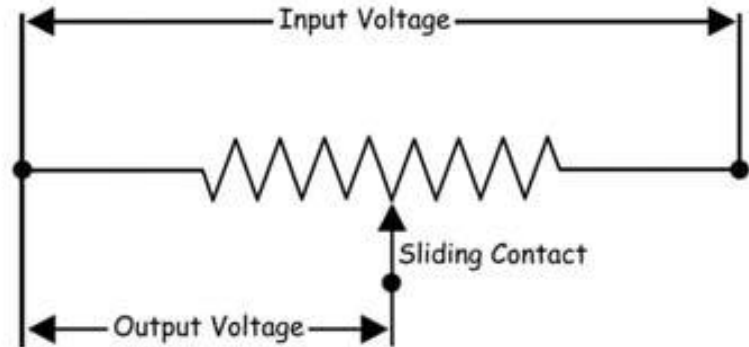
- The ITEAD studio IM130330001 2-axis joystick with pushbutton is simply **two potentiometers**, one for each axis.
 - read the voltage present on the joystick axis to provide the position of the joystick to the application
 - The select button is actuated when the joystick is pressed down.

BoosterPack Plug-in Module Header Connection	Pin Function
J1.2	Horizontal X-axis
J1.5	Select button
J3.26	Vertical Y-axis



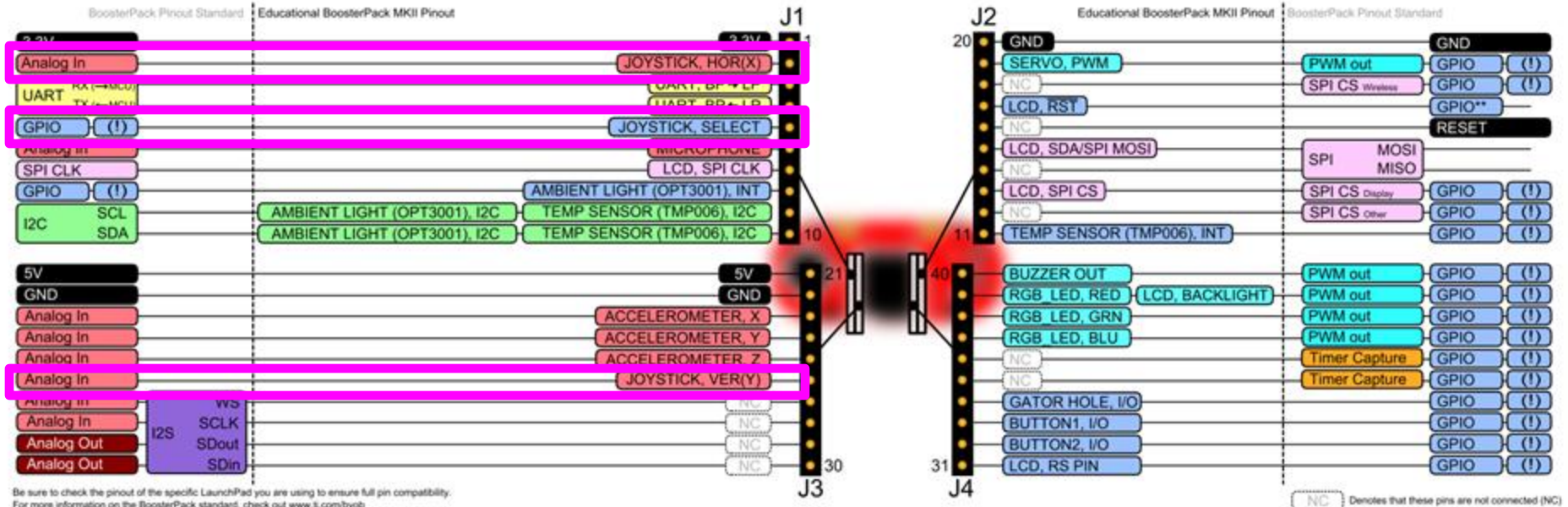
What is a Potentiometer?

- It's a resistor with a variable resistance by turning the 'wiper'.
(Connected to pin 2)
 - Position can be read as a variable analog voltage.



PinOut Diagram

- Pins **6.0** and **4.4** on launchpad are X and Y axis.



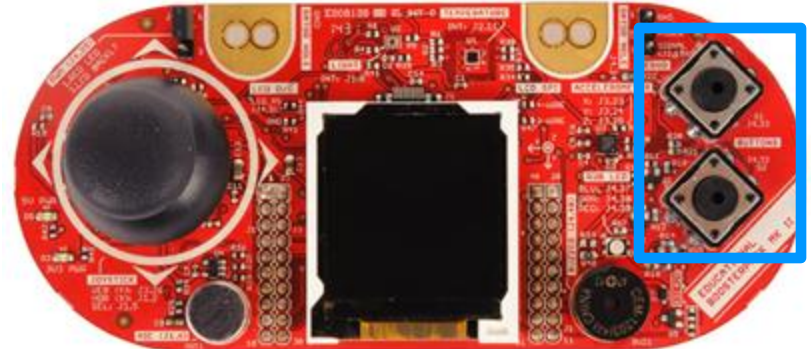
Lab #2: Moving Objects on the Screen

- Put a character on LED screen and shift it up/down and left/right based on the Joystick input.

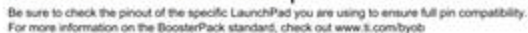
User Push Buttons

- The user pushbuttons on the BOOSTXL-EDUMKII are connected to pullup resistors that drive the BoosterPack plug-in module pin high until the button is pressed and the pin is driven low.

BoosterPack Plug-in Module Header Connection	Pin Function
J4.32	S2 button
J4.33	S1 button



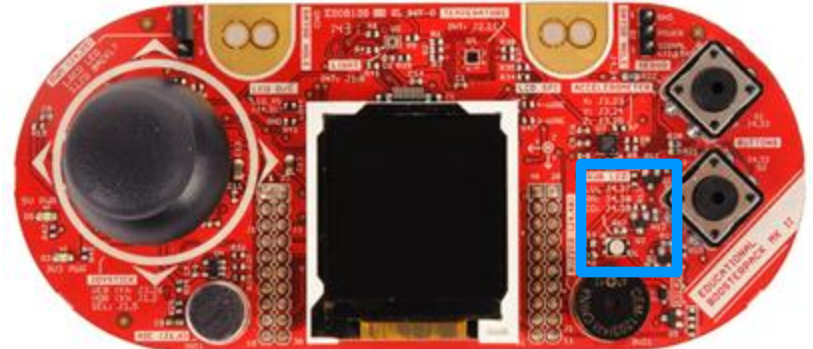
- Guess which pins?



Multicolor LED Light

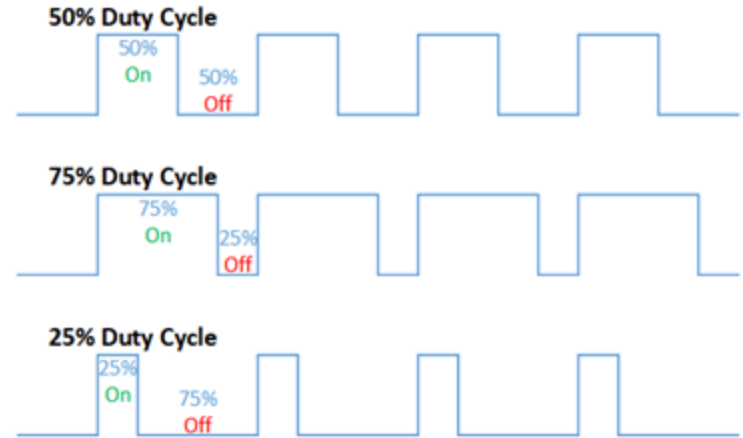
- The Cree CLV1A-FKB RGB multicolor LED light output can make any color by **mixing** red, green, and blue.
 - Each color channel can be individually modified by **pulse width modulation (PWM)** to achieve the desired color.

BoosterPack Plug-in Module Header Connection	Pin Function
J4.37	Blue channel
J4.38	Green channel
J4.39 ⁽¹⁾	Red channel



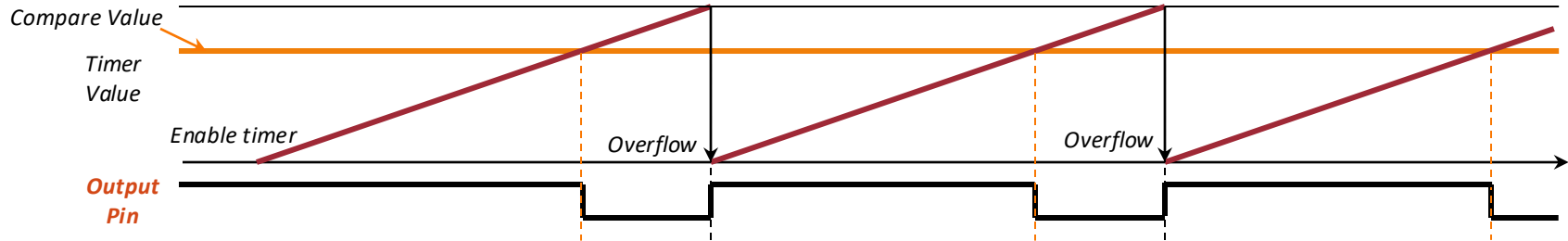
Pulse-Width Modulation

- PWM signal characteristics
 - **Modulation frequency** – how many pulses occur per second (fixed)
 - **Period** – $1/(\text{modulation frequency})$
 - **On-time** – amount of time that each pulse is on
 - **Duty-cycle** – on-time/period
- **Adjust** on-time (hence duty cycle) to represent the analog value



Timer - PWM Mode

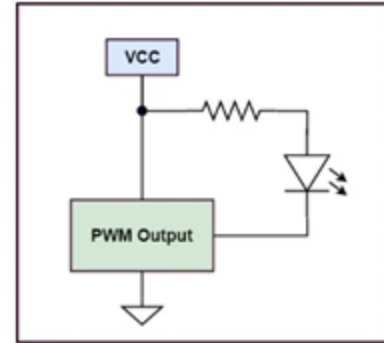
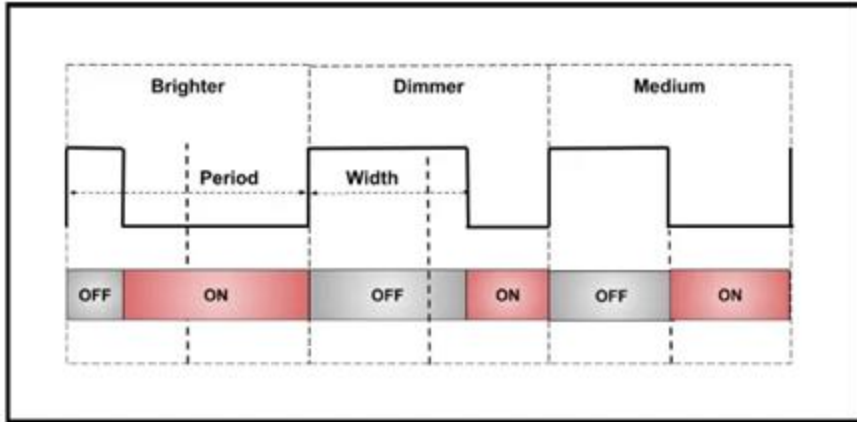
- PWM duty cycle is proportional to compare value
 - **Period:** Max timer value
 - **Pulse Width:** Compare Value



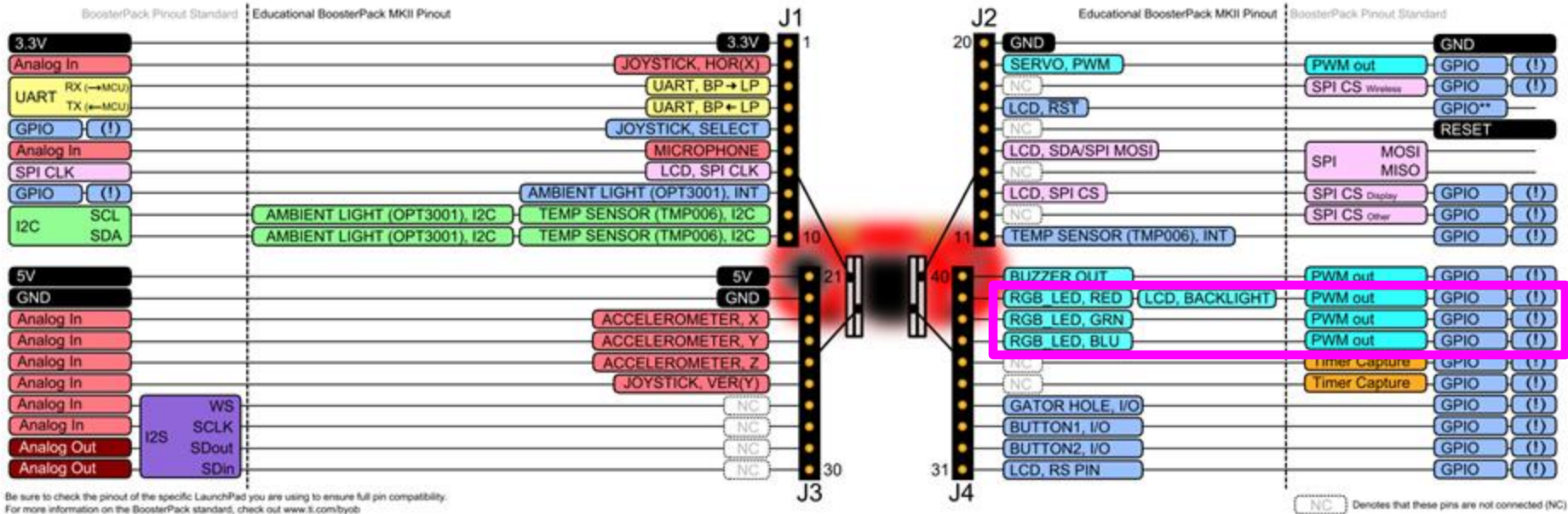
$$\text{Duty Cycle} = (\text{Compare Value} / \text{Max Value}) \times 100\%$$

Controlling LED Intensity with a PWM Timer

- LED is on when the timer output is low and off when the output is high.
 - The timer creates a rapidly changing output
 - *turns the LED on and off fast enough*
 - *the eye **averages** the “on” time and perceives a change in intensity*



PinOut Diagram



Be sure to check the pinout of the specific LaunchPad you are using to ensure full pin compatibility. For more information on the BoosterPack standard, check out www.ti.com/tvpb

In PWM Mode

- You **do not need** to **manually** toggle the GPIO pin by processing in the ISR:
 - In PWM mode, the GPIO pin is automatically toggled.
- Pins are internally connected to the timer peripheral.
 - Internally, capture-compare registers are used to adjust the duty cycle.

Controlling LED Intensity with a PWM Timer

- Pins 5.6, 2.4 and 2.6 for RGB

Timer A2 , CC1

Timer A0 , CC1

Timer A0 , CC3

Table 6-75. Port P5 (P5.6 and P5.7) Pin Functions

PIN NAME (P5.x)	x	FUNCTION	CONTROL BITS OR SIGNALS ⁽¹⁾		
			P5DIR.x	P5SEL1.x	P5SEL0.x
P5.6/TA2.1/VREF+/VeREF+/C1.7	6	P5.6 (I/O)	I: 0; O: 1	0	0
		TA2.CCI1A	0	0	1
		TA2.1	1		
		N/A	0	1	0
		DVSS	1		
P5.7/TA2.2/VREF-/VeREF-/C1.6	7	VREF+, VeREF+, C1.7 ⁽²⁾⁽³⁾	X	1	1
		P5.7 (I/O)	I: 0; O: 1	0	0
		TA2.CCI2A	0	0	1
		TA2.2	1		
		N/A	0	1	0
		DVSS	1		
		VREF-, VeREF-, C1.6 ⁽²⁾⁽³⁾	X	1	1

Table 6-67. Port P2 (P2.4 to P2.7) Pin Functions

PIN NAME (P2.x)	x	FUNCTION	CONTROL BITS OR SIGNALS ⁽¹⁾			
			P2DIR.x	P2SEL1.x	P2SEL0.x	P2MAPx
P2.4/PM_TA0.1 ⁽²⁾	4	P2.4 (I/O)	I: 0; O: 1	0	0	X
		TA0.CCI1A	0	0	1	default
		TA0.1	1			
		N/A	0	1	0	X
		DVSS	1			
		N/A	0	1	1	X
P2.5/PM_TA0.2 ⁽²⁾	5	DVSS	1			
		P2.5 (I/O)	I: 0; O: 1	0	0	X
		TA0.CCI2A	0	0	1	default
		TA0.2	1			
		N/A	0	1	0	X
		DVSS	1			
P2.6/PM_TA0.3 ⁽²⁾	6	P2.6 (I/O)	I: 0; O: 1	0	0	X
		TA0.CCI3A	0	0	1	default
		TA0.3	1			
		N/A	0	1	0	X
		DVSS	1			
		N/A	0	1	1	X
P2.7/PM_TA0.4 ⁽²⁾	7	DVSS	1			
		P2.7 (I/O)	I: 0; O: 1	0	0	X
		TA0.CCI4A	0	0	1	default
		TA0.4	1			
		N/A	0	1	0	X
		DVSS	1			
		N/A	0	1	1	X
		DVSS	1			

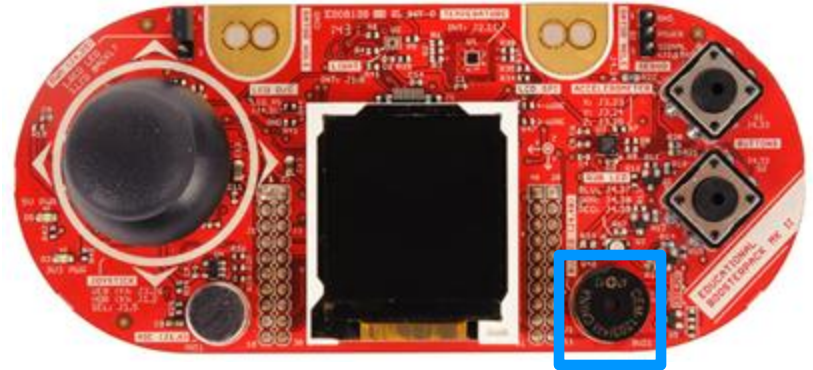
Lab#3:

- Use buttons to increase/decrease the intensity of the LEDs.

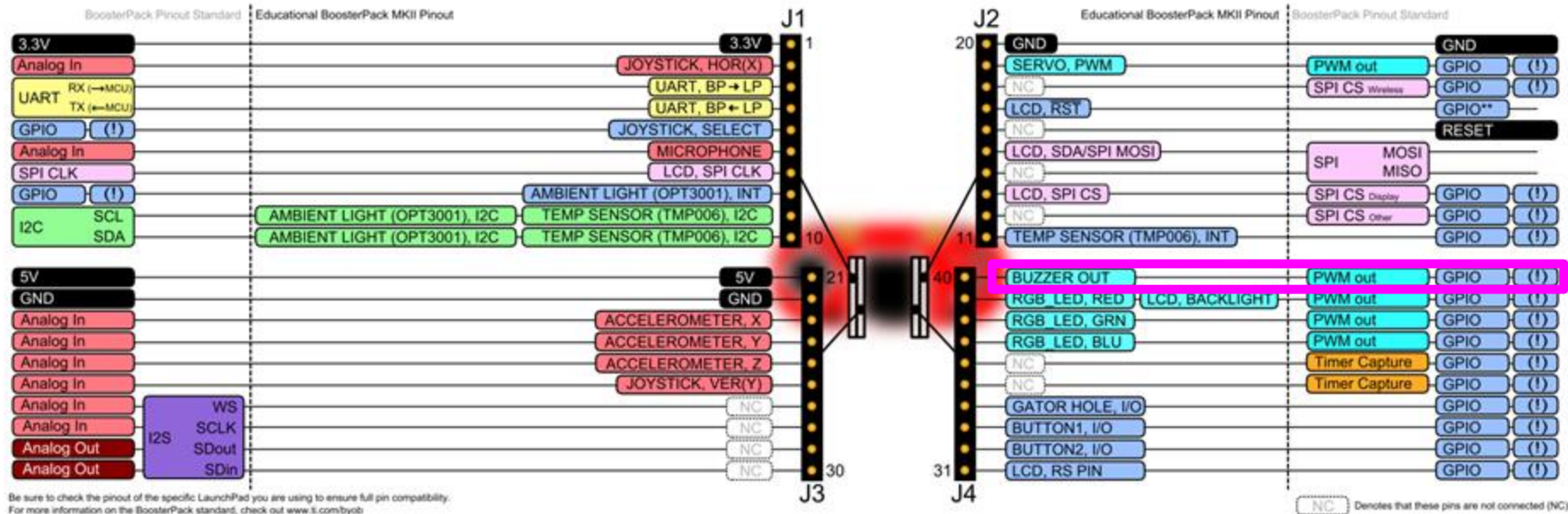
Piezo Buzzer

- The CUI CEM-1203(42) piezo buzzer can play various frequencies based on the **user-provided PWM signal**.
 - You can even play different tones back to back to create a song.

BoosterPack Plug-in Module Header Connection	Pin Function
J4.40	Buzzer input



PinOut Diagram



Be sure to check the pinout of the specific LaunchPad you are using to ensure full pin compatibility.
For more information on the BoosterPack standard, check out www.ti.com/tyotb

Piezo Buzzer

- **P2.7** is the buzzer
 - Set timer mode and adjust PWM

Table 6-67. Port P2 (P2.4 to P2.7) Pin Functions

PIN NAME (P2.x)	x	FUNCTION	CONTROL BITS OR SIGNALS ⁽¹⁾			
			P2DIR.x	P2SEL1.x	P2SEL0.x	P2MAPx
P2.4/PM_TA0.1 ⁽²⁾	4	P2.4 (I/O)	I: 0; O: 1	0	0	X
		TA0.CCI1A	0	0	1	default
		TA0.1	1			
		N/A	0	1	0	X
		DVSS	1			
		N/A	0			
		DVSS	1			
P2.5/PM_TA0.2 ⁽²⁾	5	P2.5 (I/O)	I: 0; O: 1	0	0	X
		TA0.CCI2A	0	0	1	default
		TA0.2	1			
		N/A	0	1	0	X
		DVSS	1			
		N/A	0			
		DVSS	1			
P2.6/PM_TA0.3 ⁽²⁾	6	P2.6 (I/O)	I: 0; O: 1	0	0	X
		TA0.CCI3A	0	0	1	default
		TA0.3	1			
		N/A	0	1	0	X
		DVSS	1			
		N/A	0			
		DVSS	1			
P2.7/PM_TA0.4 ⁽²⁾	7	P2.7 (I/O)	I: 0; O: 1	0	0	X
		TA0.CCI4A	0	0	1	default
		TA0.4	1			
		N/A	0	1	0	X
		DVSS	1			
		N/A	0			
		DVSS	1			

Timer A0, CC4

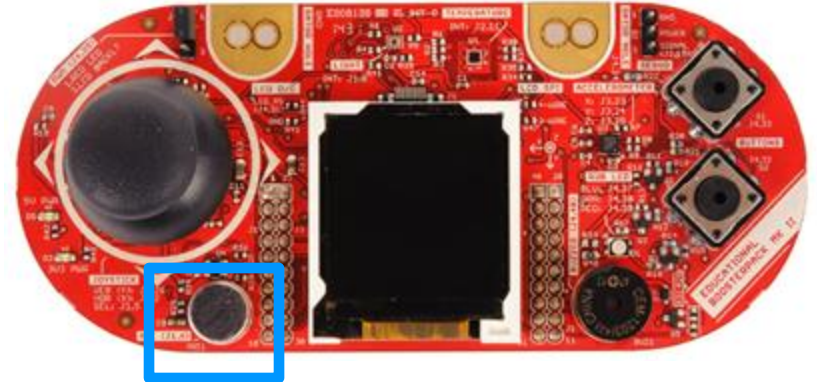
Lab #4:

- Try to produce some sounds via the buzzer
 - Buzzer will emit a sound according to the frequency and amplitude of vibration.
 - It allows to play notes and to recreate simple melodies.
- What do we need to do to play music with Buzzer?

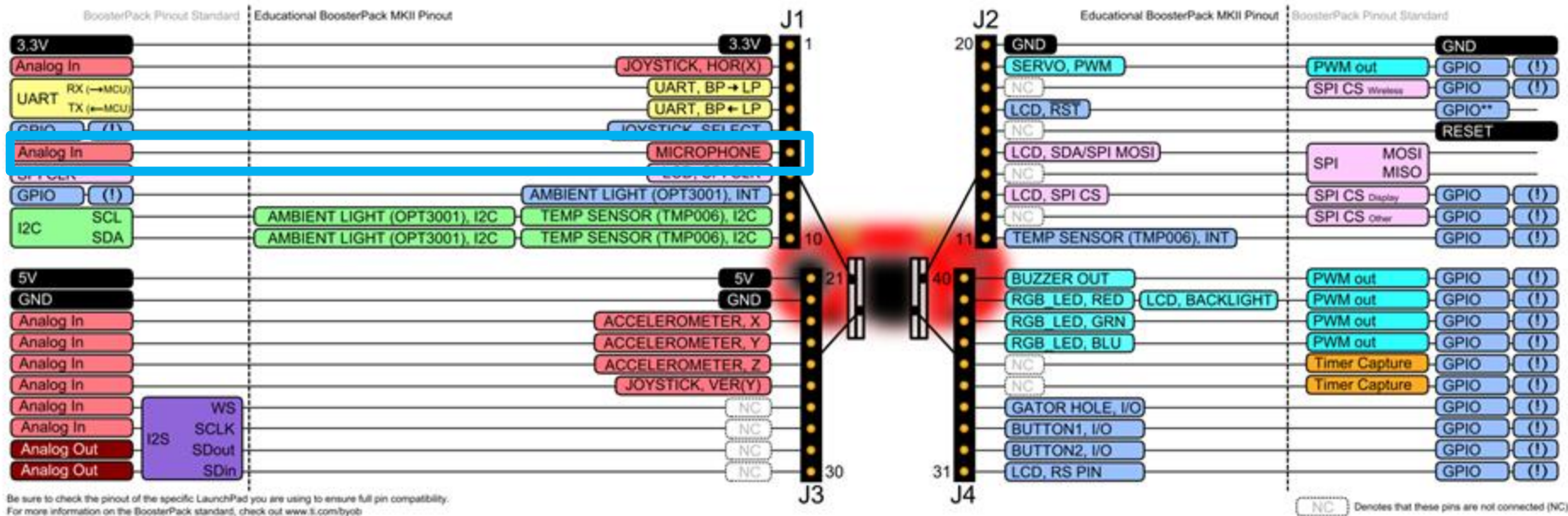
Microphone

- The CUI CMA-4544PF-W microphone uses an OPA344 operational amplifier to boost the output of the microphone.
 - The human ear can hear frequencies between 0 and 20 kHz and the operating range of the microphone is 20 Hz to 20 kHz.

BoosterPack Plug-in Module Header Connection	Pin Function
J1.6	Microphone Output



- **Pin 4.3** on launchpad

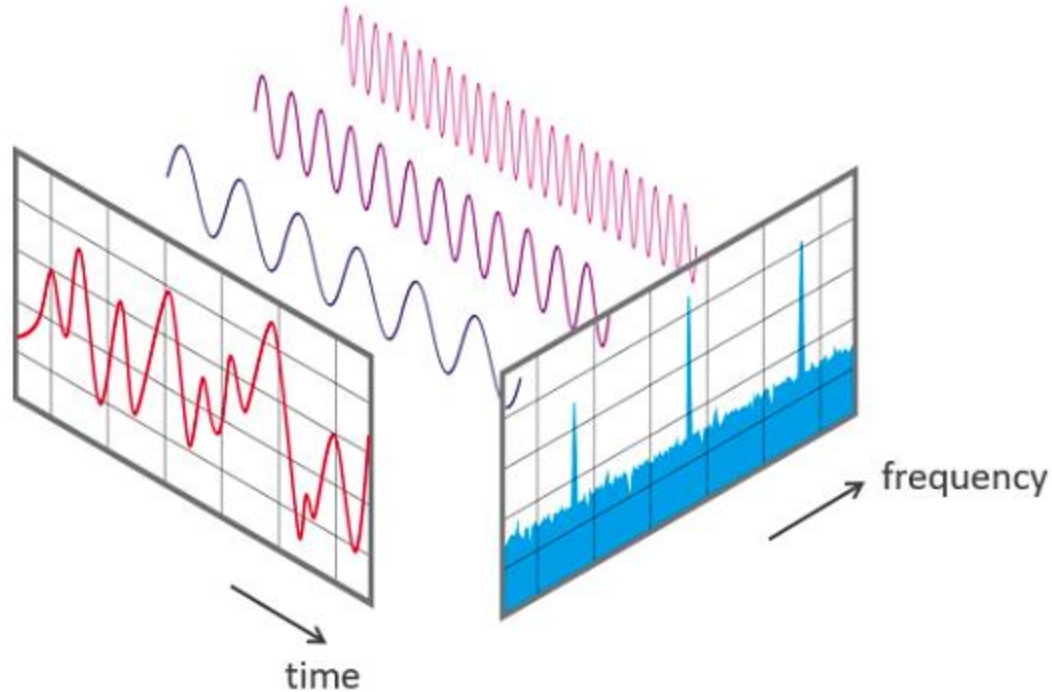


Lab #5: Detect Noise!

- Write a program that will blink an LED in a noisy environment.

FFT (Fast Fourier Transform)

- A transform takes one signal and turns it into another signal
 - From time domain to the frequency domain
- More details:
https://www.youtube.com/watch?v=h7apO7q16V0&ab_channel=Reducible

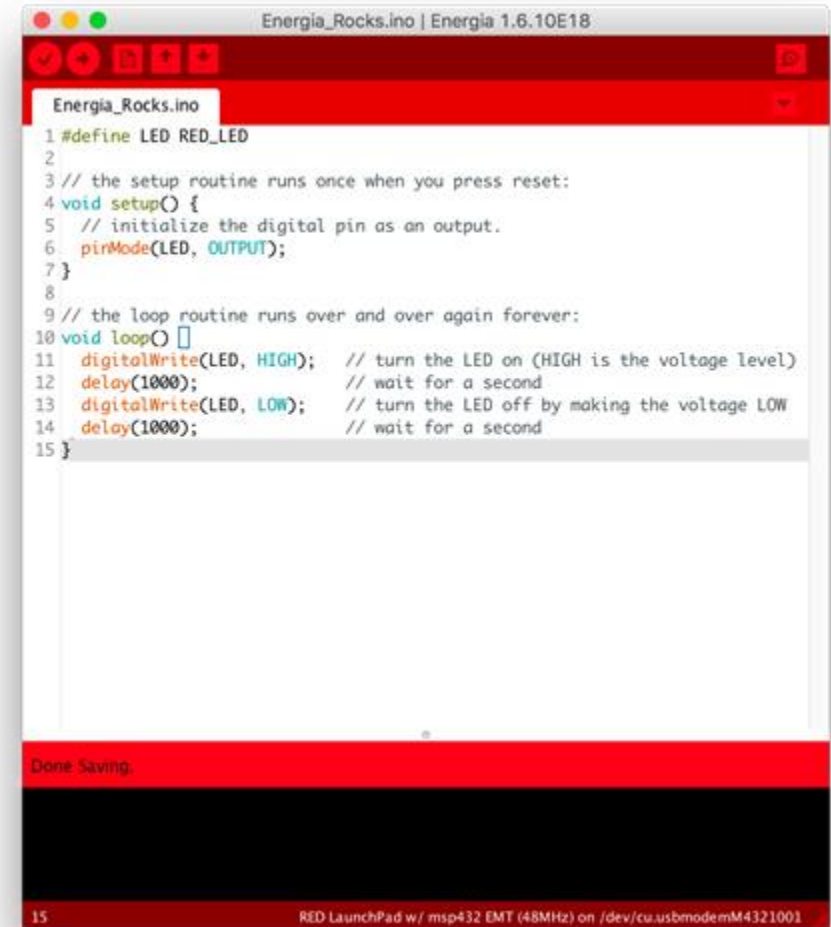


Lab #6: Check FFT Example with the Microphone.

- How can we use the frequencies to understand activities?

Energia

- <https://energia.nu/>
 - Brings the Wiring and Arduino framework to the Texas Instruments MSP432 LaunchPad.
- You can check and see if it works
 - Feel free to use it too.



The screenshot shows the Energia IDE interface. The title bar reads "Energia_Rocks.ino | Energia 1.6.10E18". The code editor displays the following C++ code:

```
1 #define LED RED_LED
2
3 // the setup routine runs once when you press reset:
4 void setup() {
5   // initialize the digital pin as an output.
6   pinMode(LED, OUTPUT);
7 }
8
9 // the loop routine runs over and over again forever:
10 void loop() {
11   digitalWrite(LED, HIGH); // turn the LED on (HIGH is the voltage level)
12   delay(1000);             // wait for a second
13   digitalWrite(LED, LOW);  // turn the LED off by making the voltage LOW
14   delay(1000);             // wait for a second
15 }
```

Below the code editor, a red status bar displays the message "Done Saving." At the very bottom, a black status bar shows the page number "15" and the hardware configuration: "RED LaunchPad w/ msp432 EMT (48MHz) on /dev/cu.usbmodemM4321001".